

Mohamed Ali

Senior AI (GenAI) Engineer | AI Architect

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SUMMARY

Senior AI Engineer and Architect with 7+ years of experience in Generative AI, Multi-Agent Systems, MLOps, and enterprise-scale AI architecture. Proven track record delivering production-grade solutions in logistics automation, fraud detection, predictive analytics, computer vision, and advanced NLP/RAG systems. Expert in designing microservices-based AI platforms with secure deployment pipelines for mission-critical business applications.

PROFESSIONAL EXPERIENCE

Senior AI (GenAI) Engineer | AI Architect, Norconsult Telematics

Oct 2025 – Present

Full-time · Riyadh, Saudi Arabia · On-site

- Built an **AI-Powered Truck Safety Inspection System** for checkpoint gates: face verification (InsightFace SCRFD + ArcFace), bilingual Arabic/English license OCR (PaddleOCR), and license-to-face cross-verification via 512-d embeddings — achieving **96.7% detection accuracy at 56 ms/image on CPU** and sub-350 ms end-to-end inspection, eliminating manual identity checks entirely.
- Benchmarked 6 face-detection models across 3,226 WIDER Face images (61 scene categories) and 4 OCR engines; designed a privacy-first architecture storing only non-reversible embeddings.
- Architected and deployed a centralized **Multi-Agent GenAI Operations Platform** integrating 4 AI modules (Anomaly Detection, Delay Prediction, OCR, Agentic Chatbot) using a microservices “Sidecar Gateway” pattern with Nginx, FastAPI, Streamlit, and Docker — serving 5+ departments.
- Engineered an **Autonomous Agentic Chatbot** with RAG capabilities using GPT-4, LangChain, and ChromaDB/Pinecone over structured (SQL/CSV) and unstructured logistics data, reducing manual query time by **40%** across 200+ daily queries.
- Developed an **Intelligent Anomaly Detection System** using ensemble ML (Isolation Forests + statistical heuristics) to flag fraud patterns across 10,000+ daily supply-chain waybills, improving detection rate by **35%**.
- Built a **Predictive Logistics Delay Engine** with XGBoost/LightGBM and 30+ engineered features (route congestion, contractor history, seasonality), reaching **87% accuracy** and cutting scheduling conflicts by 25%.
- Implemented an **Enterprise Document Intelligence & OCR** pipeline (PaddleOCR/Tesseract with GPT-4 post-processing and Human-in-the-Loop validation), reducing document processing time by **60%**.
- Established secure production infrastructure with zero-trust tunneling (zrok), systemd self-healing orchestration, and RBAC enforcement via cryptographically signed sessions.

Confidential

Jul 2023 – Feb 2026 · 2 yrs 8 mos

Full-time · 2 positions · promoted once

Senior Data Scientist | Senior AI Engineer **PROMOTION**

Jul 2024 – Feb 2026

Riyadh, Saudi Arabia · Remote

Focus: Large Language Models (LLM)

- Architected an **AI Voice-First Interview Platform** automating candidate screening end-to-end: a LangGraph stateful conversation engine, real-time Faster Whisper STT → LLM correction → Coqui XTTS v2 TTS pipeline, multi-LLM scoring (GPT-4o + Ollama), and CV-based behavioral analysis (MediaPipe, InsightFace). Delivered 50+ REST/WebSocket APIs, 13+ data models, and 127+ unit tests with sub-3 s turn latency.
- Architected and developed **WeBook Brain**, an integrated AI system featuring a Gemini 2-powered dual-agent chatbot, a high-performance Fraud Detection RAG application, and a recommendation system — consolidating 3 separate systems into 1 platform and halving support response time.
- Engineered a scalable FastAPI **fraud detection backend** using Ollama 3, Mistral, ChromaDB, and MongoDB, containerized with Docker for cross-platform deployment across 100K+ historical transactions.
- Developed a robust **churn prediction model** using gradient boosting ensembles and 50+ engineered behavioral features, achieving **89% accuracy** with 30-day advance warning for retention campaigns.

AI Engineer | Data Scientist

Jul 2023 – Jun 2024

Egypt · Hybrid
Focus: Generative AI

- Built and implemented advanced predictive models for sales forecasting, customer segmentation, fraud prevention, and churn prediction to drive strategic business decisions.
- Optimized product demand predictions by deploying ML and DL models including ARIMA, SARIMA, and Prophet, leading to improved strategic planning.
- Extracted actionable insights from complex datasets to refine marketing campaigns and enhance overall decision-making processes.

Machine Learning and Operations Engineer, MDS for Computer Systems (MDS CS)

May 2022 – Jul 2023

Full-time · Saudi Arabia · Hybrid

- Designed, developed, and deployed scalable AI systems utilizing large-scale, real-world datasets for industrial research and prediction algorithms.
- Constructed a deep learning model to accurately predict network traffic load for large-scale network devices in a real infrastructure environment and successfully deployed it into production.

Data Scientist | AI Engineer, Linked Gates

May 2019 – Apr 2021

Full-time · Cairo, Egypt · On-site

- Constructed intelligent software systems using the Odoo framework, integrated with AI models to deliver predictive insights for client-specific use cases.
- Developed an image classification system using Convolutional Neural Networks (CNNs) trained on the ImageNet dataset for high-accuracy categorization.
- Engineered a comprehensive restaurant management platform (VOO) with a mobile application to streamline order management, inventory, and customer engagement.

EDUCATION

M.Sc., Computing and Artificial Intelligence

2022 – 2025

Fayoum University, Egypt

B.Sc., Computing and Artificial Intelligence

2015 – 2019

Fayoum University, Egypt

PUBLICATIONS

Video Captioning using Deep Learning with Greedy Search (VCDLGS)

<https://doi.org/10.1016/j.fraope.2026.100497> — Elsevier

KEY PROJECTS

PRODUCTION SYSTEMS

AI-Powered Offline Asset Intelligence Platform (2026) – Norconsult Telematics

Offline AI platform for data-center engineering drawings. Extracts and reconciles equipment tags using entity resolution (Fellegi-Sunter, CRF, Belief Propagation), predictive ML with active learning, and explainable AI. Single CPU-only executable with zero cloud dependency.

AI Truck Safety Inspection System (2025–2026) – Norconsult Telematics

Driver verification pipeline with 96.7% face detection accuracy at 56 ms on CPU, bilingual license OCR, and license-to-face cross-verification. Benchmarked 6 detection and 4 OCR models with interactive Plotly analytics.

AI Voice-First Interview Platform (2025–2026)

Production GenAI platform with LangGraph conversational agents, real-time STT/TTS, multi-LLM scoring (GPT-4o + Ollama), and computer vision behavioral analysis. 50+ APIs, sub-3 s latency, 70% reduction in recruiter workload.

Multi-Agent GenAI Operations Platform (2025–2026) – Norconsult Telematics

Centralized GenAI platform modernizing enterprise dispatch and logistics operations. Integrated 4 AI modules using a microservices “Sidecar Gateway” pattern with Nginx, FastAPI, and Streamlit. Chain-of-Thought prompting for complex multi-step logistics queries.

Intelligent Anomaly Detection System (2025–2026) – Norconsult Telematics

Real-time fraud detection using ensemble ML (Isolation Forests + statistical heuristics). React/Vite SPA frontend with FastAPI backend for interactive “Deep Dive” dashboards over supply-chain waybills.

Predictive Logistics Delay Engine (2025–2026) – Norconsult Telematics

XGBoost/LightGBM models for delivery delay forecasting with engineered features (route congestion, contractor history, seasonality). Streamlit dashboard visualizing Delay Severity probabilities with actionable recommendations.

Enterprise Document Intelligence & OCR (2025–2026) – Norconsult Telematics

Multi-stage OCR pipeline using PaddleOCR with LLM post-processing (GPT-4) for error correction and JSON structuring. Human-in-the-Loop validation UI for low-confidence extractions. Reduced processing time by 60%.

WeBook Assistant (2025)

Dual-agent AI assistant based on Gemini 2, combining data API and chatbot for seamless support.

Fraud RAG (2025)

Fraud detection app using Ollama 3, Mistral, ChromaDB, and MongoDB with a scalable FastAPI backend.

Network Traffic Load Prediction (2022–2023) – MDS for Computer Systems

Deep learning model forecasting device load in network infrastructure environments, deployed to production.

VOO Platform (2019–2020) – Linked Gates

Restaurant management platform and mobile app for streamlined order and inventory control.

RESEARCH & EXPERIMENTS

Multi-Lingual RAG (2025)

RAG system for processing and generating content across multiple languages.

Real-Time Fraud Detection (2025)

High-throughput, low-latency system for real-time transaction analysis to prevent financial loss.

Bias Detection in AI Models (2025)

Framework to identify and mitigate biases in ML models, ensuring fairness and ethical AI performance.

Anxiety Detection (2024)

Predictive model detecting early signs of anxiety using physiological and behavioral data.

ECG Classification (2024)

Deep learning model for automated ECG signal classification to identify cardiac arrhythmias.

Mini-RAG (2024)

Lightweight RAG application using Mistral LLM and txtai for semantic search.

Brain Tumor Segmentation (2024)

YOLOX-based X-ray system for hyper-resolution brain tumor segmentation and classification.

Churn Prediction (2024)

ML pipeline using advanced feature engineering to improve customer retention.

Smart Robot (2024)

AI model for solar panel cleaning robots, integrating dust sensors for efficiency optimization.

Sales Forecasting (2023)

ARIMA, SARIMA, and Prophet models for best-in-class sales prediction.

Fatigue Prediction (2023)

Hybrid learning pipeline (CatBoost, XGBoost, LightGBM, deep NN) for fatigue detection.

Arabic OCR (2022)

Arabic text extraction systems using Docker, Flask, and REST API for document digitization.

Video Captioning (2021–2023) PUBLISHED

Research using ResNet, VGG, LSTM, and GloVe to generate natural captions for video content.

Image Classification System (2020)

CNN-based system trained on ImageNet for high-accuracy multi-category classification.

SCADA Innovation (2020)

Automated manufacturing software using Odoo and Postgres for high-precision production.

TECHNICAL SKILLS

Programming: Python, R, C++, Bash Scripting

Generative AI: LLM Fine-tuning (Qwen 3/Llama/Mistral), Prompt Engineering, Agentic Workflows, RAG, Chain-of-Thought Prompting, LangChain, LangGraph, OpenAI API, Hugging Face Transformers

ML/DL: TensorFlow, PyTorch, Scikit-Learn, XGBoost, LightGBM, CatBoost, Isolation Forests, Anomaly Detection, Time-Series Forecasting (ARIMA/SARIMA/Prophet), OpenCV, MLflow, NumPy

Computer Vision & NLP: OCR (Tesseract/PaddleOCR), Document Intelligence, CNN, ResNet, YOLOX, InsightFace, MediaPipe, Natural Language Processing

Vector Databases: ChromaDB, FAISS, Qdrant, Pinecone

Big Data: PySpark, Pandas, Dask

MLOps & DevOps: Docker, Kubernetes, GitHub Actions, AWS SageMaker, Azure ML, CI/CD, DVC, Systemd, Nginx, Reverse Proxy, Microservices Architecture

Backend & APIs: FastAPI, Flask, REST API, WebSockets, Odoo

Frontend: React (Vite), Streamlit, Gradio

Databases: MongoDB, PostgreSQL, SQL, ChromaDB

BI/Visualization: Power BI, Plotly, Matplotlib, Seaborn, Interactive Dashboards

AI Security: Responsible AI, Data Governance & Compliance, RBAC, Zero-Trust Networking

SOFT SKILLS

- Problem Solving & Critical Thinking
- Technical Leadership & Architecture Design
- Cross-Functional Team Collaboration
- Agile/Scrum Methodologies
- Stakeholder Communication

HONORS

- 1st Place, Pre-master Class, Fayoum University (2022)
- 1st Place, Graduation Projects, Fayoum University (2019)
- 3rd Place, IEEE-ICPC Programming Competition (2018)

VOLUNTEER EXPERIENCE

- GDG Fayoum
- IEEE-FSB
- Life Makers

LANGUAGES

- **Arabic:** Native
- **English:** Full Professional